

P16-M Monitor Mixer

Musician Training Guide

FOR	Worship Team Musicians
PURPOSE	Learn to control your personal monitor mix
DEVICE	Behringer Powerplay P16-M
PRACTICE	p16-simulator.cloud
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CONTENTS

01	What is the P16-M?
02	Channel Layout for Your Team
03	Controls — Knobs & Faders Explained
04	Workflow: Dialing In Your Mix
05	Hearing Safety
06	Real-World Scenarios
07	Common Mistakes to Avoid
08	Knowledge Check Quiz
09	Quick Reference Card

MODULE 01

What is the P16-M?

The Behringer Powerplay P16-M is a **personal in-ear monitor mixer**. It lets each musician on stage control their own headphone mix — independently from the main FOH (front of house) mix.

Instead of asking the sound engineer to "turn up the drums in my ears," you control your own blend right on this unit. Your adjustments **do not affect the main speakers** or any other musician's monitor mix.

How it fits in your church setup

- 1
- The main console sends all 16 channels via a single AES50 / Ultranet cable to a P16-D distributor.
- 2
- The distributor feeds up to 8 P16-M units — one for each musician (drummer, keys player, guitarist, etc.).
- 3
- Each musician plugs their in-ear monitors or headphones into their P16-M and controls their own mix.
- 4
- Your mix changes do NOT affect the main speaker sound or anyone else's monitor mix.

Pro tip:
When you first sit down, set all channels to 0 and start fresh. Slowly bring up your own instrument first, then add what you need for timing (drums/click), then blend in the rest.

MODULE 02

Channel Layout for Your Team

All 16 channels are pre-assigned to specific instruments in your church setup. Knowing which channel is which lets you dial in your mix quickly during soundcheck.

Important:
Channel assignments shown here may differ between campuses. This layout needs adjustment for each campus — always confirm with your campus sound team which instruments are routed to which channels before your first rehearsal.

CH 01 DRUM L Left overhead / drum room mic. Pan left for stereo image.	CH 02 DRUM R Right overhead / room mic. Pair with Ch.01, pan right.
CH 03 KEYS L Left output of keyboard/piano. Pan slightly left.	CH 04 KEYS R Right output of keyboard/piano. Pan slightly right.

MODULE 03

Controls — Knobs & Faders Explained

BASSLow Frequency EQ | ± 12 dB @ 100 Hz, shelving

Shelving EQ fixed at 100 Hz with up to 12 dB of boost or cut. Affects all frequencies below 100 Hz equally — like a shelf. Boost if your mix feels thin and lacks warmth. Cut if it sounds boomy or muddy. Small adjustments go a long way.

MIDMidrange EQ | ± 12 dB, semi-parametric

Semi-parametric midrange EQ with up to 12 dB of boost or cut. Use together with the FREQ knob to target a specific midrange frequency. This is where vocals and instruments sit — a slight cut can reduce harshness; boosting too much makes the mix sound "honky."

FREQ

Mid Frequency Sweep | 100 Hz to 10 kHz, adjustable

Sweeps the center frequency of the MID band across the full range of 100 Hz to 10 kHz. Turn FREQ while boosting MID to find the problem frequency, then cut it back. This is the most surgical EQ tool on the unit — use it to zero in on harshness or muddiness.

TREBLEHigh Frequency EQ | ± 12 dB @ 10 kHz, shelving

Shelving EQ fixed at 10 kHz with up to 12 dB of boost or cut. Affects all frequencies above 10 kHz equally. Boost for more "air" and clarity on cymbals or acoustic guitar. Cut if your mix is too bright or sibilant (harsh "s" sounds in vocals).

LIMITERPeak Protection | Ratio 1: ∞

Hard-limiting with an infinite ratio (1: ∞) — once the signal hits the threshold, it is clamped completely, preventing any peaks above that level. Sets a hard ceiling on the maximum output volume. Set this before every service to protect your hearing from sudden loud spikes.

LEVEL

Output Level

Sets the output level going to your IEM or headphones. This is separate from master VOLUME. Use it to trim if one side feels louder than the other.

PAN/BAL

Stereo Position

For stereo channel pairs (like DRUM L/R or KEYS L/R), pan each channel left or right. For mono sources, center is usually best unless you want spatial separation.

VOLUME

Master Volume

The big knob. Controls your total mix loudness. Start at 50% and adjust. Never max this out — protect your hearing, especially during a long service.

CHANNEL FADERS

Individual Channel Volume

The vertical sliders at the bottom. Each controls how loud one specific instrument is in your headphones. This is your primary mixing tool — use it the most.

STORE

Save User Preset

Saves your current mix settings to one of the 16 available user presets. Use this after dialing in a mix you are happy with so you can recall it instantly at the next rehearsal or service without starting from scratch.

RECALL

Load User Preset

Loads one of the 16 saved user presets. If you have previously stored your mix with the STORE button, press RECALL to bring it back instantly. Great for switching between different service configurations quickly.

GROUP

Channel Grouping

Groups any selected mono or stereo-linked channels into one or more groups, which can then be stored into user presets. Useful when you want to control multiple related channels together — for example, grouping all drum channels so you can adjust them as one unit.

LINK

Stereo Channel Linking

Combines adjacent mono channels into linked stereo pairs. For example, linking Ch.01 and Ch.02 creates a stereo drum pair that behaves as one stereo channel. Use this for any source that is natively stereo, such as drums, keys, or backing tracks.

MODULE 04

Workflow: Dialing In Your Mix

- 1** **Select a channel** — Press its number button at the bottom of the unit.
- 2** **Adjust the channel fader** — The vertical slider sets that instrument's volume in your ears.
- 3** **Set Pan/Balance** — For stereo pairs (drum L/R, keys L/R), pan them left and right respectively.
- 4** **Use EQ sparingly** — Bass/Mid/Treble are global. Only adjust if the overall sound is too harsh or muddy.
- 5** **Set master VOLUME** — Start at 50% and increase slowly. Keep under 75% to protect your hearing.
- 6** **Set the LIMITER** — Set a safe ceiling before the service starts — treat this as a non-negotiable step.

MODULE 05

Hearing Safety

Hearing warning:

In-ear monitors deliver sound directly into your ear canal. Long services at high volume cause permanent hearing damage. Use the LIMITER knob to set a ceiling, and take the volume down if you feel ringing after rehearsal.

Volume limit: Keep master VOLUME below 75%. Louder does not mean better — it means faster hearing damage.

Use the LIMITER: Set the LIMITER knob before every service. It acts as a safety ceiling for sudden peaks.

Take breaks: During long rehearsals, remove your IEMs for a few minutes every hour.

Watch for ringing: If your ears ring after rehearsal, that is a warning sign. Reduce your volume next time.

Fit matters: A good IEM seal actually lets you use lower volumes. Invest in well-fitting earpieces.

MODULE 06

Real-World Scenarios

Here are common situations you'll face and how to fix them on the P16-M:

Can't hear the click track

Problem: You keep losing the beat because the click is buried.

Solution: Press channel 15 (CLICK). Raise the channel fader. If still buried, slightly lower all other channels so the click cuts through. The click should be clearly audible without being painfully loud.

IEMs sound too harsh

Problem: The mix is uncomfortably bright — cymbals and vocals are piercing.

Solution: Press the MAIN button, then go to EQ and slightly reduce TREBLE. If that's not enough, use MID and FREQ together: sweep FREQ while cutting MID to pinpoint and reduce the harsh frequency.

Can't hear yourself playing

Problem: Your instrument is completely lost in the mix.

Solution: Press your channel (e.g., Ch.09 GTR1) and raise the fader significantly. Then slightly reduce channels you don't need as much to make room in your headphone mix.

Mix is too quiet overall

Problem: Even with VOLUME up, everything sounds too soft.

Solution: Press the MAIN button to check and adjust the main volume level. Also verify your IEM cable is fully plugged in. Check the LEVEL knob (output trim) and raise it if needed. Do NOT raise individual channels to max — that causes distortion.

No sound at all

Problem: You plug in but hear absolutely nothing.

Solution: Check in order: (1) Is the Ultranet/AES50 cable connected? (2) Is VOLUME above 0? (3) Are your IEMs fully plugged in? (4) Is the unit powered on — the ULTRANET indicator should be green.

MODULE 07

Common Mistakes to Avoid

Maxing the VOLUME knob:

This can damage your hearing and cause distortion. Start at 50% and work up.

Boosting all EQ bands:

Boosting bass, mid, and treble together just raises the volume — not clarity. Cut what is bad rather than boosting everything.

Pulling a mic channel too high:

Can create uncomfortable resonance. Keep mic channels (especially overhead/room mics) moderate.

Skipping soundcheck:

Always dial in your mix before the service with your IEMs physically in place. The mix sounds different in your ears vs. open air.

Not resetting between users:

If someone else used your unit, always start from scratch. Their mix is for their ears, not yours.

Ignoring the LIMITER:

Set it every time. It takes 5 seconds and protects your hearing for a lifetime.

MODULE 08

Knowledge Check Quiz

Answer these questions to confirm your understanding. Correct answers are marked with a checkmark.

1. Which control adjusts the treble frequencies for your entire mix?

- A) Channel fader
- ✓ B) TREBLE knob in the EQ section
- C) Master VOLUME knob
- D) LIMITER knob

Explanation: The TREBLE knob in the EQ section affects high frequencies across your whole mix. Channel faders only control one channel's volume.

2. A musician says "I need the engineer to turn up the keys in my ears." What should actually happen?

- A) The FOH engineer adjusts the main mix
- ✓ B) The musician adjusts channels 3 & 4 on their P16-M
- C) The musician raises the master VOLUME
- D) The musician presses SOLO

Explanation: The whole point of the P16-M is personal control. Channels 3 (KEYS L) and 4 (KEYS R) are the keyboard channels.

3. What does the LIMITER knob do?

- A) Mutes all channels simultaneously
- ✓ **B) Sets a maximum volume ceiling to protect your hearing**
- C) Boosts all bass frequencies
- D) Controls click track volume

Explanation: The LIMITER sets a peak output ceiling. Set it before each service so sudden loud sounds cannot damage your hearing.

4. The click track is on which channel?

- A) Channel 8
- B) Channel 13
- ✓ **C) Channel 15**
- D) Channel 16

Explanation: Channel 15 is labeled CLICK. Drummers and bass players especially should keep this clearly audible.

5. Your mix sounds boomy and muddy. Which EQ knob do you reduce?

- A) TREBLE
- B) MID
- C) FREQ
- ✓ **D) BASS**

Explanation: The BASS knob controls low frequencies. A muddy mix has too much bass — a small reduction cleans it up.

MODULE 09

Quick Reference Card

Cut this page out and keep it on your monitor wedge stand or music stand for quick access.

CONTROL	WHAT IT DOES	WHEN TO TOUCH IT
BASS knob	Boosts/cuts low frequencies	Mix sounds boomy (cut) or thin (boost)
MID knob	Boosts/cuts midrange	Mix sounds harsh or honky
FREQ knob	Sweeps mid frequency	Use with MID to find harsh freq
TREBLE knob	Boosts/cuts high freq	Mix too bright (cut) or dull (boost)
LIMITER	Sets max volume ceiling	Set BEFORE every service
LEVEL	Output trim	One side louder than other
PAN/BAL	Stereo positioning	Stereo pairs like drums/keys
VOLUME	Master loudness	Overall mix too quiet/loud
Channel fader	Individual channel volume	Any time — this is your main tool
CH 15 (CLICK)	Metronome click track	Raise when you lose the beat
CH 13 (ALL VCL)	All vocal mics blend	Need to hear melody/lyrics

No sound? Check in order:

1. Ultramet/AES50 cable is connected to the unit
2. VOLUME knob is above 0
3. IEMs/headphones are fully plugged in
4. Unit is powered on (ULTRAMET light = green)
5. At least one channel fader is raised above 0

Pro tip:

Practice on the simulator at p16-simulator.cloud before your next rehearsal. Explore each channel and knob until the layout feels natural.

This training guide was created for the worship team. For questions, speak with your music director.

Simulator: p16-simulator.cloud | **Device:** Behringer Powerplay P16-M

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